MATH 345 Skeleton Notes
Unit 1: Vectors, matrices, and linear maps
Section 1.1: Vectors

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Vectors as lists of numbers

A real quantity is also called a scalar.

Definition: Vector

An n-vector is an ordered list of n scalars, often written in the form $\mathbf{v} = (v_1, v_2, \dots, v_n)$. The collection of all n-vectors is denoted $\mathbb{R}^n$.

We will also encounter vectors written as a row or column of numbers in square brackets, i.e.,

$$\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}.$$

It is customary to use bold letters for variables, such as $\mathbf{v}$ or $\mathbf{w}$, to represent vectors. We let $\mathbf{0}$ denote the vector $(0, \dots, 0)$ consisting only of zeros. Two vectors $\mathbf{v}$ and $\mathbf{w}$ are equal if they have the same number of entries and all the corresponding entries are equal. A point in n-dimensional space can also be identified with an ordered list of n numbers using Cartesian coordinates. When referring specifically to a point P, we write $P(x_1, x_2, \dots, x_n)$ to indicate its coordinates.

Vectors as displacements in space: building intuition

Vectors can be used to represent displacements between points in space. The vector $\mathbf{v} = (v_1, v_2, v_3)$ represents the displacement from $P(x_1, y_1, z_1)$ to $Q(x_2, y_2, z_2)$, where $x_2 = x_1 + v_1$, $y_2 = y_1 + v_2$, and $z_2 = z_1 + v_3$. This vector is also written as $\overrightarrow{PQ}$. Because vectors often indicate displacements, they are drawn pictorially as arrows starting at a point, and ending where that point is displaced by the vector.

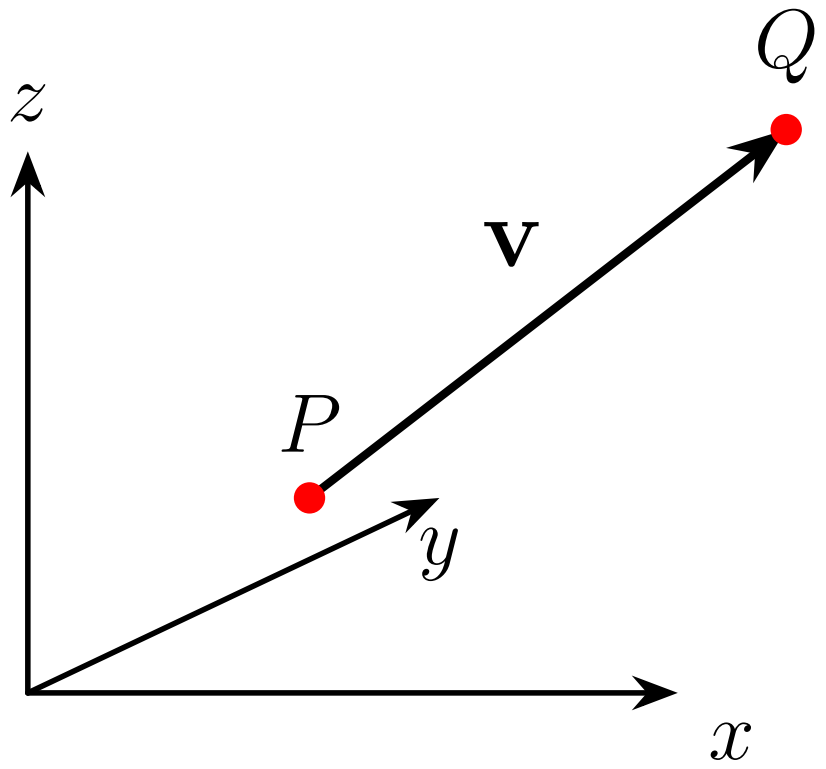


Figure. A vector $\mathbf{v}$ represented as an arrow between two points P and Q.

Activity: Vectors as displacements in space (U1-LO2)

What is the vector representing the displacement from the point $P(1, 2)$ to the point $Q(5, 7)$.

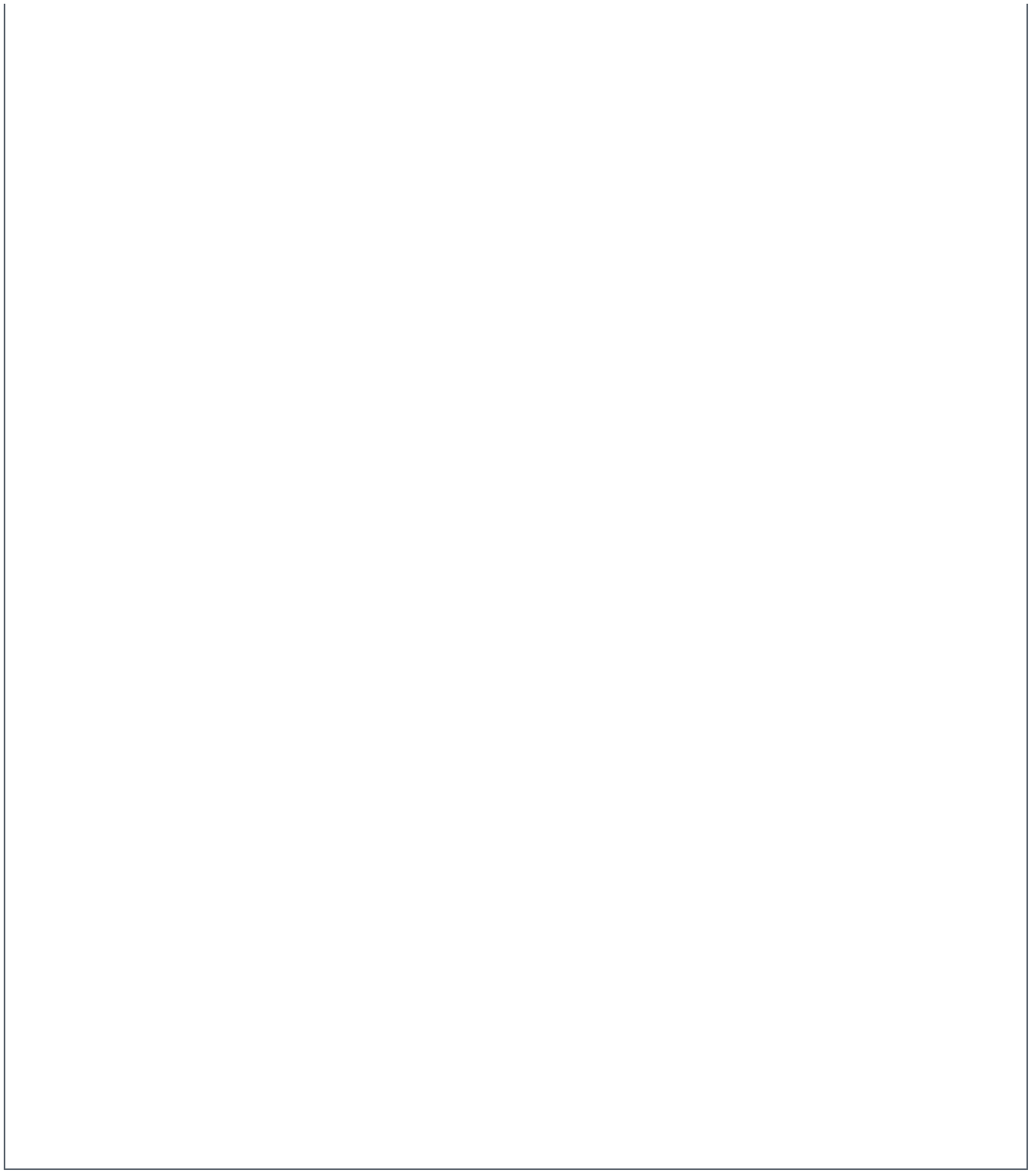
Definition: Length of a vector

The length of a vector $\mathbf{v} = (v_1, \ldots, v_n)$, also called its Euclidean norm, is the quantity

$$\|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}.$$

Activity: Vector length and geometry (U1-LO1)

Verify, using planar geometry, that the length of a 2-vector is the length of the line-segment of the arrow representing the vector.



In a calculus course, you may have mostly seen a vector as a displacement, but in this course it can also be a row of data or a list of model parameters, among other important applications. The same operations below will later compare documents, tokens, and feature vectors.

Adding and scaling vectors

Two displacements can be combined: just apply one displacement after the other is applied. This naturally leads to the notion of *adding* two vectors.

Definition: Adding vectors

The sum of two n-vectors $\mathbf{v} = (v_1, \dots, v_n)$ and $\mathbf{w} = (w_1, \dots, w_n)$, denoted $\mathbf{v} + \mathbf{w}$, is the vector given by the tuple $(v_1 + w_1, \dots, v_n + w_n)$.

Remark

The sum of two vectors can be illustrated pictorially as in the figure by drawing the parallelogram with the vectors placed along two sides of the parallelogram.

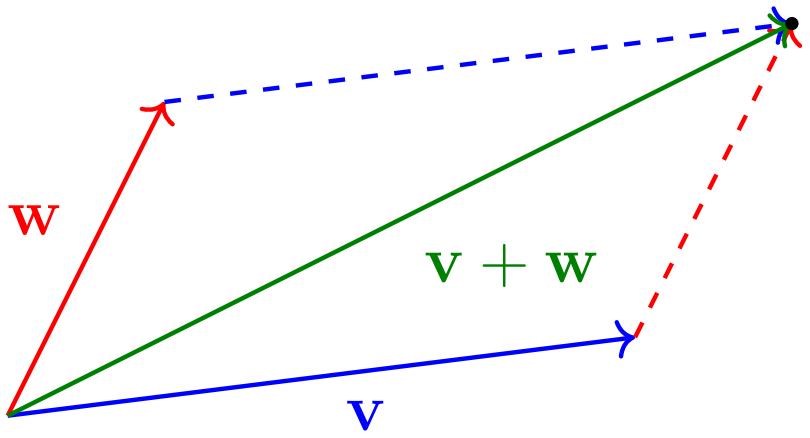


Figure. The sum of two vectors $\mathbf{v}$ and $\mathbf{w}$.

One can also *scale* a vector by a given scalar quantity.

Definition: Scaling vectors

If $\mathbf{v} = (v_1, \cdots, v_n)$ is an n-vector and $t \in \mathbb{R}$ is a scalar, we let $t\mathbf{v}$ denote the vector $(tv_1, \cdots, tv_n)$, i.e., multiplying each entry of the vector $\mathbf{v}$ by t.

Definition: Unit vectors and normalization

A unit vector is a vector of length one. If $\mathbf{x}$ is nonzero,

the unit vector in the direction of $\mathbf{x}$ is

$$\mathbf{u} = \frac{\mathbf{x}}{\|\mathbf{x}\|}.$$

Dividing a nonzero vector by its norm is called normalization.
It preserves the vector's direction and gives it length one.

Definition: Linear combinations and convex combinations of vectors

If $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are vectors in $\mathbb{R}^n$ and $c_1, \ldots, c_k$ are scalars, then

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$$

is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$. The scalars $c_1, \ldots, c_k$ are the coefficients of the linear combination.
If $c_i \geq 0$ for all i and $c_1 + \cdots + c_k = 1$, then the linear combination is a convex combination.
A convex combination can be interpreted as a weighted average.

Activity: A weighted average (U1-LO8)

Let $\mathbf{v}_1 = (10, 0)$, $\mathbf{v}_2 = (0, 10)$, and $\mathbf{v}_3 = (10, 10)$. Let $\alpha_1 = 1/4$, $\alpha_2 = 1/4$, and $\alpha_3 = 1/2$. Compute $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3$.

Example: Subtracting vectors as a linear combination

Subtracting a vector means adding its negative:

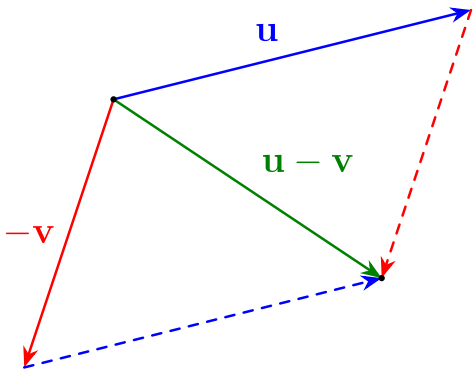
$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}) = 1\mathbf{u} + (-1)\mathbf{v}.$$

Thus, $\mathbf{u} - \mathbf{v}$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$ with coefficients 1 and -1. For example, if $\mathbf{u} = (4, 1)$ and $\mathbf{v} = (1, 3)$, then

$$\mathbf{u} - \mathbf{v} = (4, 1) + (-1, -3) = (3, -2).$$

Geometrically, $-\mathbf{v}$ has the same length as $\mathbf{v}$ but points in the opposite direction. We can therefore use the parallelogram construction from the figure with $\mathbf{u}$ and $-\mathbf{v}$, as shown on the left in the figure. Equivalently, when $\mathbf{u}$ and $\mathbf{v}$ start at the same point, the arrow from the tip of $\mathbf{v}$ to the tip of $\mathbf{u}$ represents $\mathbf{u} - \mathbf{v}$. Here, it is the displacement from $P(1, 3)$ to $Q(4, 1)$, shown on the right.

(a) Add the negative vector



(b) Join the tips

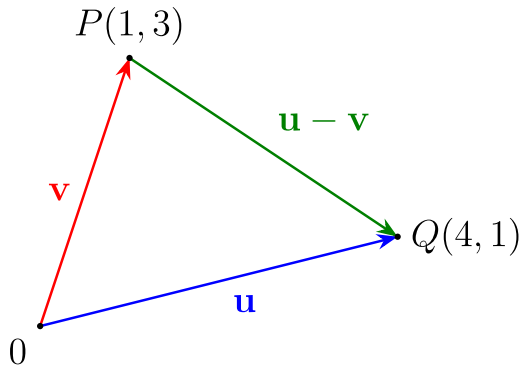


Figure. The difference $\mathbf{u} - \mathbf{v}$ as the sum $\mathbf{u} + (-\mathbf{v})$ (left) and as the displacement from the tip of $\mathbf{v}$ to the tip of $\mathbf{u}$ (right).

Dot products of vectors

Definition: Dot product

The dot product of two n-vectors $\mathbf{v} = (v_1, \ldots, v_n)$ and $\mathbf{w} = (w_1, \ldots, w_n)$ is the scalar quantity

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \cdots + v_n w_n.$$

Theorem: Properties of the dot product

Let $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ be vectors in $\mathbb{R}^n$, and let a be a scalar in $\mathbb{R}$. Then the following properties hold:

- $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.
- $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$.
- $(a\mathbf{x}) \cdot \mathbf{y} = a(\mathbf{x} \cdot \mathbf{y}) = \mathbf{x} \cdot (a\mathbf{y})$.
- $\mathbf{x} \cdot \mathbf{x} \geq 0$, and $\mathbf{x} \cdot \mathbf{x} = 0$ if and only if $\mathbf{x} = \mathbf{0}$.
- $\|a\mathbf{x}\| = |a|\|\mathbf{x}\|$.

Note

Note that $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$ for any $\mathbf{v} \in \mathbb{R}^n$.

Activity: Expanding the squared norm of a sum (U1-LO1)

Show that $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2$ for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Definition: Orthogonal vectors

Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are

orthogonal, or perpendicular, if

$$\mathbf{x} \cdot \mathbf{y} = 0.$$

This definition applies to all vectors, including the zero vector, which is orthogonal to every vector.

Geometrically, the dot product of two vectors is a quantity which is related to the *angle* between two nonzero vectors $\mathbf{v}$ and $\mathbf{w}$. If we draw the two vectors as arrows with the same starting point, then they form an angle on the plane containing both vectors, and the angle θ is given below.

Theorem: Angle between vectors

For two nonzero vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, the angle θ between the vectors is the quantity satisfying the equation

$$\cos\theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|}.$$

Theorem: The Cauchy-Schwarz inequality

If $\mathbf{x}, \mathbf{y}$ are vectors in $\mathbb{R}^n$,

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|.$$

Equality holds if either vector is zero. For nonzero vectors, equality holds if and only if they are scalar multiples of each other.

The dot product can be thought of as a measure of the *similarity* between two vectors. Suppose θ is the angle between two nonzero vectors $\mathbf{v}$ and $\mathbf{w}$:

- If θ is close to zero, then the two vectors are close to pointing in the same direction. Since $\cos(0^\circ) = 1$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} \approx 1.$$

- If θ is close to 180°, then the two vectors are close to pointing in opposite directions. Since $\cos(180^\circ) = -1$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} \approx -1.$$

- If θ is close to 90°, the two vectors are close to pointing at right angles. Since $\cos(90^\circ) = 0$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} \approx 0.$$

  For nonzero vectors, orthogonality therefore means that the angle between them is 90°.

The cosine similarity between the nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ is denoted by

$$\text{cosim}(\mathbf{v}, \mathbf{w}) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|}.$$

It ranges between -1 and 1, and measures the degree to which the two vectors point in the same, or opposite, directions. This terminology is especially common in certain areas of data science.

Warning

Cosine similarity is undefined if either vector is the zero vector.

Activity: Range and interpretation of cosine similarity (U1-LO1)

Let $\mathbf{u}$ and $\mathbf{v}$ be nonzero vectors in $\mathbb{R}^n$. Their cosine similarity is

$$\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}.$$

1. Use the Cauchy-Schwarz inequality to explain why this number is between -1 and 1.
2. What condition makes the cosine similarity equal to 0, and what does this say geometrically?

Activity: Computing cosine similarity (U1-LO1)

Calculate the cosine similarity between the two vectors $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (\sqrt{3} + 2, 2\sqrt{3} - 1)$.

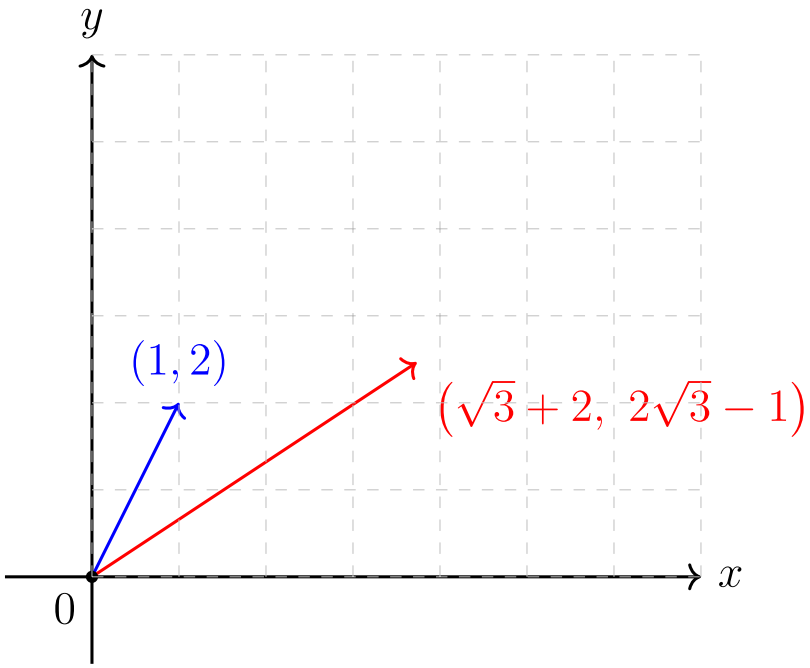


Figure. The vectors $\mathbf{v}$ and $\mathbf{w}$ from Computing cosine similarity (U1-LO1), drawn as arrows from the origin.

Activity: Finding the angle between vectors (U1-LO1)

Calculate the angle between the two vectors

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}.$$

We will study the dot product in far more detail in Orthogonality and least squares and Abstract vector spaces and polynomial approximation.

Definition: Distance between vectors

The Euclidean distance between two vectors $\mathbf{u} = (u_1, \dots, u_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$ in $\mathbb{R}^n$ is

$$\operatorname{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + \cdots + (u_n - v_n)^2}.$$

The same formula gives the distance between two points with these coordinates.

As illustrated in the figure, $\mathbf{u} - \mathbf{v}$ points from the tip of $\mathbf{v}$ to the tip of $\mathbf{u}$ when the two vectors start at the same point. Its Euclidean norm therefore measures how far apart those endpoints are. For the vectors in Subtracting vectors as a linear combination,

$$\operatorname{dist}(\mathbf{u}, \mathbf{v}) = \|(3, -2)\| = \sqrt{3^2 + (-2)^2} = \sqrt{13}.$$

In contrast, cosine similarity compares direction after normalization.

Corollary: The triangle inequality

If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, then

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

Remark

The triangle inequality for vectors implies the triangle inequality for the distance function, i.e., that

$$\operatorname{dist}(\mathbf{x}, \mathbf{z}) \leq \operatorname{dist}(\mathbf{x}, \mathbf{y}) + \operatorname{dist}(\mathbf{y}, \mathbf{z}).$$

Indeed, we calculate that

$$\begin{aligned} \operatorname{dist}(\mathbf{x}, \mathbf{z}) &= \|\mathbf{x} - \mathbf{z}\| \\ &= \|\mathbf{x} - \mathbf{y} + \mathbf{y} - \mathbf{z}\| \\ &\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\| \\ &= \operatorname{dist}(\mathbf{x}, \mathbf{y}) + \operatorname{dist}(\mathbf{y}, \mathbf{z}). \end{aligned}$$

Applications of vectors

Vectors can represent quantities other than displacements. The entries of a vector can record measurements, counts, samples, or features. The meaning of a vector depends on what its coordinates represent.

- **Object:** Color
  Meaning of the entries: red, green, and blue intensities
- **Object:** Time series
  Meaning of the entries: measurements at successive times
- **Object:** Portfolio
  Meaning of the entries: amounts held in each asset

- **Object:** Image
  Meaning of the entries: pixel intensities, listed in a fixed order
- **Object:** Document
  Meaning of the entries: counts of selected words
- **Object:** Customer
  Meaning of the entries: purchases of selected products
- **Object:** Object with features
  Meaning of the entries: measured attributes such as size, price, weight, or rating

The same vector operations can have different interpretations. A sum can add purchases, add word counts, or add two time series. A scalar multiple can rescale an image, double a portfolio, or change units. A dot product can produce a score. A distance can compare two feature vectors. Cosine similarity compares direction after normalization.

Activity: Word-count vectors (U1-LO1, U1-LO2)

Use the dictionary `linear`, `matrix`, `data`. A document vector records the number of times these words appear, in this order. The query

$$\mathbf{q} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

represents the phrase “linear data”. Consider three document vectors

$$\mathbf{D}_1 = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}, \qquad \mathbf{D}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{D}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

Compute the cosine similarities and Euclidean distances between $\mathbf{q}$ and each $\mathbf{D}_i$. Rank the documents from most to least similar by cosine similarity and from closest to farthest by Euclidean distance.

In the previous activity, $\mathbf{D}_1$ is most similar to the query by cosine similarity but farthest from it by Euclidean distance; $\mathbf{D}_2$ is closest by Euclidean distance. The two measures select different documents because cosine similarity compares direction after normalization, while Euclidean distance also depends on vector length.

Activity: The same rankings in code (U1-LO2, U1-LO8)

The following code repeats the cosine-similarity and Euclidean-distance rankings from Word-count vectors (U1-LO1, U1-LO2). We compute the cosine similarities between the query and each document, then the Euclidean distances by taking the norm of each difference.

```
import numpy as np

D1 = np.array([3., 0., 3.])
D2 = np.array([1., 1., 0.])
D3 = np.array([0., 2., 0.])
q = np.array([1., 0., 1.])

score1 = (D1 @ q) / (np.linalg.norm(D1) * np.linalg.norm(q))
score2 = (D2 @ q) / (np.linalg.norm(D2) * np.linalg.norm(q))
score3 = (D3 @ q) / (np.linalg.norm(D3) * np.linalg.norm(q))

distance1 = np.linalg.norm(D1 - q)
distance2 = np.linalg.norm(D2 - q)
```

Activity: The same rankings in code (U1-LO2, U1-LO8) (continued)

```
distance3 = np.linalg.norm(D3 - q)

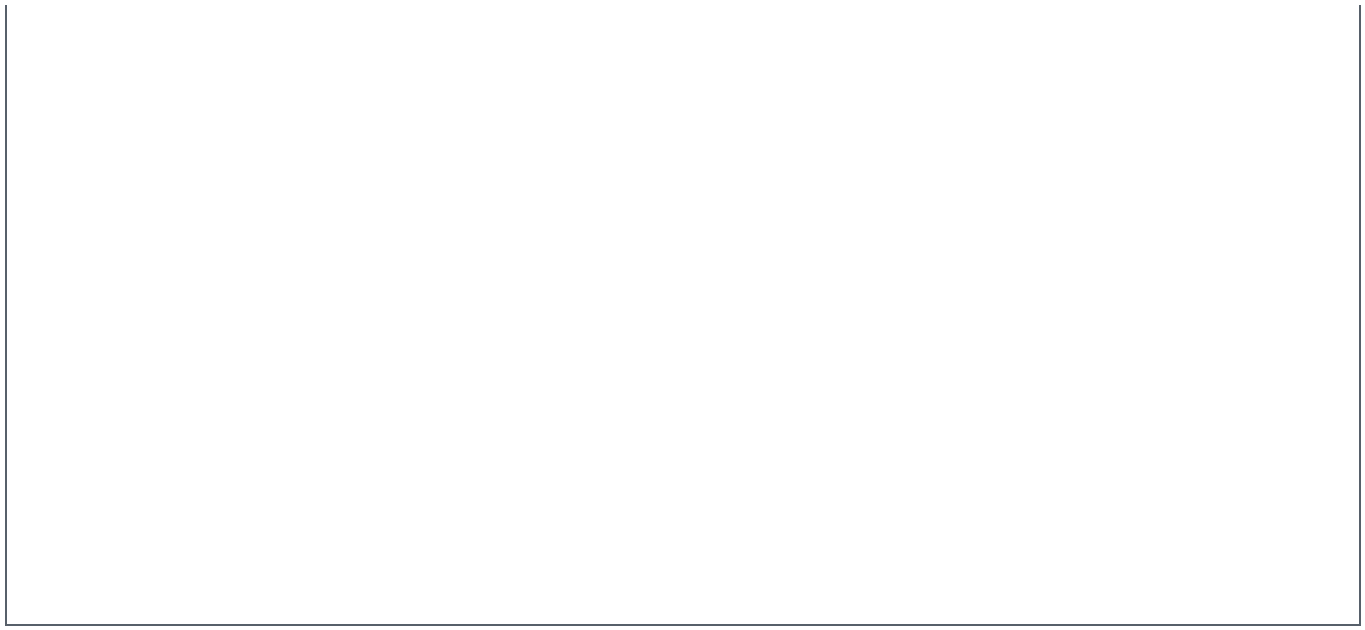
(score1, score2, score3), (distance1, distance2, distance3)
```

Output (approximately):

```
((1.0, 0.5, 0.0),
 (2.8284271247461903, 1.4142135623730951, 2.449489742783178))
```

The first tuple gives the cosine similarities and the second gives the Euclidean distances, each in the order $\mathbf{D}_1$, $\mathbf{D}_2$, $\mathbf{D}_3$ relative to $\mathbf{q}$. Smaller distances mean closer documents.

1. Which score corresponds to $\mathbf{D}_2$?
2. Where does the code compute $\mathbf{D}_1 \cdot \mathbf{q}$?
3. Which document is most similar to the query by cosine similarity?
4. Which line computes $\|\mathbf{D}_2 - \mathbf{q}\|$?
5. Rank the documents from closest to farthest from the query by Euclidean distance. How does this ranking compare with the cosine-similarity ranking?



Example: Neural networks and transformers

A neural network is a function built from layers. A layer takes numbers as input, combines them using weights, applies a rule, and sends output numbers to the next layer. In many neural networks, the input and output of a layer are vectors. The weights are adjusted from data during training.

A transformer is a neural-network architecture for sequences. In a language model, text is first broken into tokens. A token can be a word, part of a word, punctuation mark, or other text fragment. Each token position carries a vector.

A transformer layer produces a new vector at each token position. The token itself does not change; its vector becomes context-dependent. In the phrase “small red bird”, the new vector at the bird position can gather information from “small” and “red”. In a next-token model, the vector at this final position can then help predict what comes next.

For one output position, its query is used to decide which positions are relevant. Each token has a key used in that compar-

ison and a value containing the information it can contribute. Queries and keys determine the weights; values are what get averaged. Every position has all three roles, but the next activity computes only the attention output at the bird position.

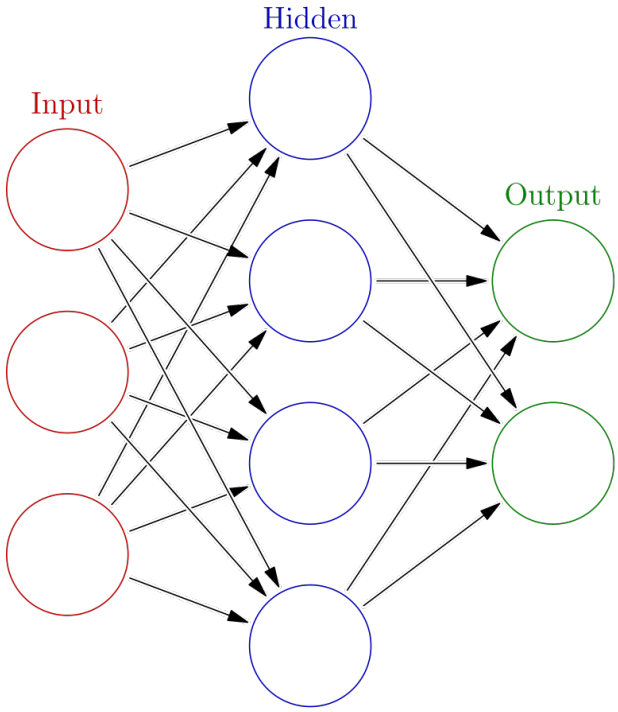


Figure. A neural network diagram with input, hidden, and output layers. Source: Wikipedia, Neural network (machine learning), using Colored neural network.svg by Glosser.ca, derivative of Artificial neural network.svg, licensed under CC BY-SA 3.0.

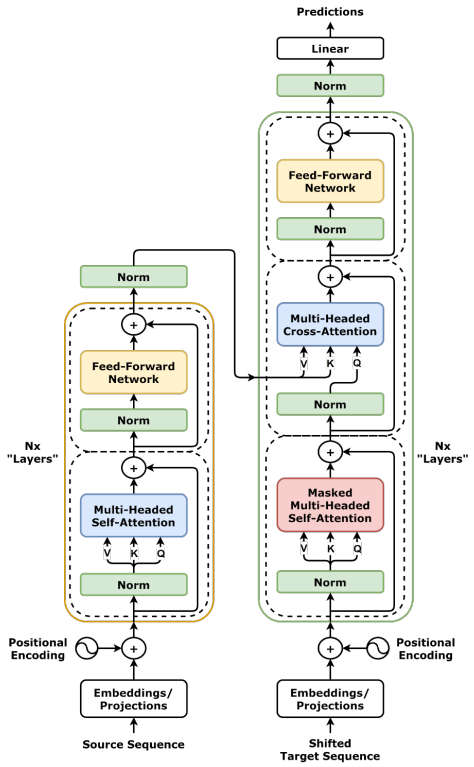


Figure. A transformer processes a sequence of token vectors. The activities below focus on the vector operations visible in the diagram: token embeddings, dot-product scores, weighted averages, and affine layers. Source: Wikipedia, Transformer (deep learning architecture), using Transformer, full architecture.png by dvgodoy, licensed under CC BY 4.0.

Activity: Gathering context for the bird position (U1-LO1, U1-LO8)

Consider the token sequence

small, red, bird.

We compute the attention output at the position occupied by “bird”. The bird position is the destination; “small”, “red”, and “bird” itself are possible sources of information.

The query $\mathbf{q}_{\text{bird}}$ is compared with each key $\mathbf{k}_i$ to produce a relevance score. The scores are converted into weights, and those weights are used to average the value vectors:

$$\text{query–key scores} \longrightarrow \text{weights} \longrightarrow \text{weighted average of values.}$$

Activity: Gathering context for the bird position (U1-LO1, U1-LO8) (continued)

For this activity, the score for token i is

$$\text{score}_i = \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_i.$$

Use

$$\mathbf{q}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{k}_{\text{small}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{k}_{\text{red}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{k}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the three scores

$$\mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{small}}, \qquad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{red}}, \qquad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{bird}}.$$

2. Normalize the positive scores by dividing each score by the sum of all three scores. Call the resulting weights

$$\alpha = \begin{bmatrix} \alpha_{\text{small}} \\ \alpha_{\text{red}} \\ \alpha_{\text{bird}} \end{bmatrix}.$$

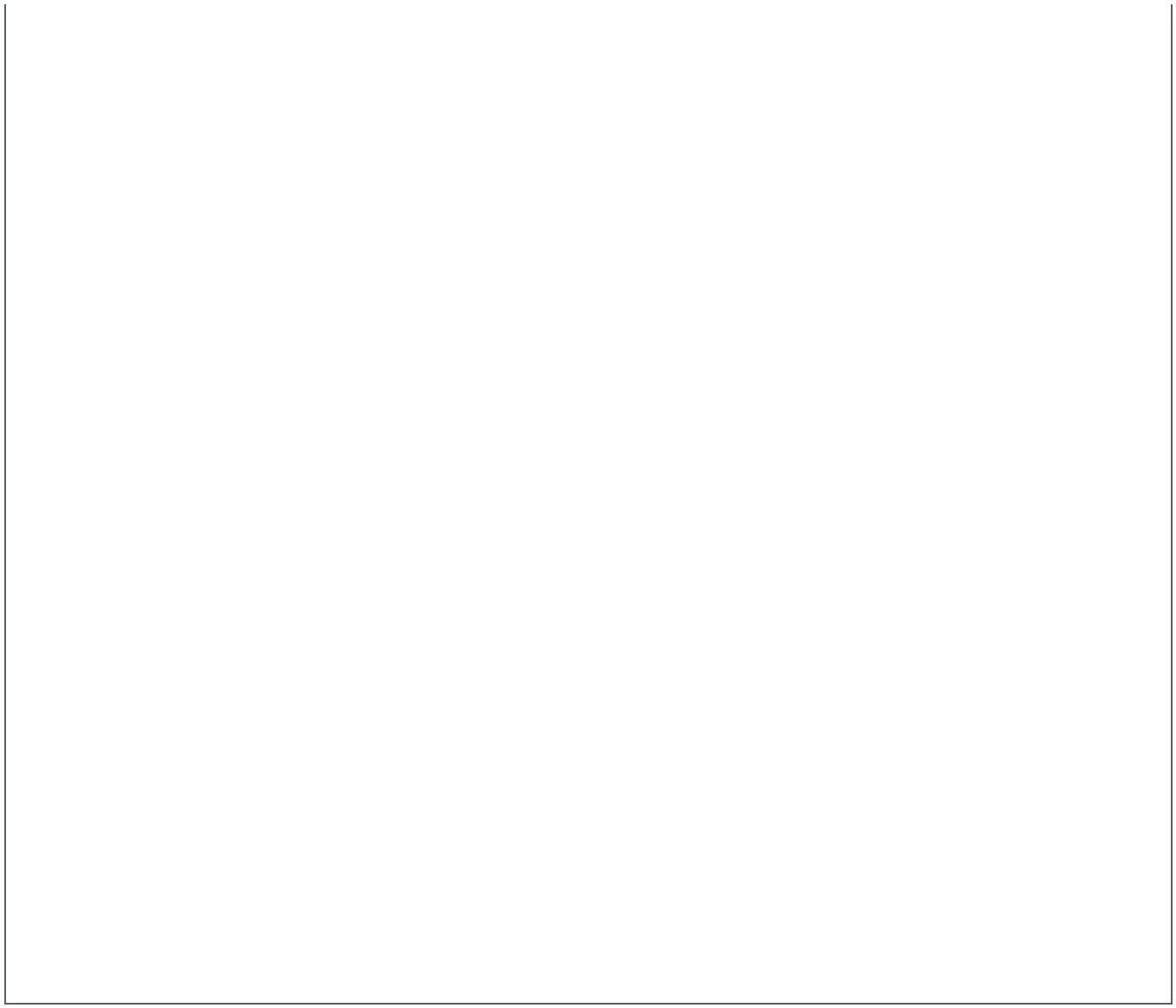
3. The coordinates of the value vectors are abstract toy features. Use

$$\mathbf{v}_{\text{small}} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_{\text{red}} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \qquad \mathbf{v}_{\text{bird}} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

to compute

$$\mathbf{h}_{\text{bird}} = \alpha_{\text{small}} \mathbf{v}_{\text{small}} + \alpha_{\text{red}} \mathbf{v}_{\text{red}} + \alpha_{\text{bird}} \mathbf{v}_{\text{bird}}.$$

4. Which token receives the largest weight? What total weight is assigned to “small” and “red” together? What does this say about the information gathered at the bird position?



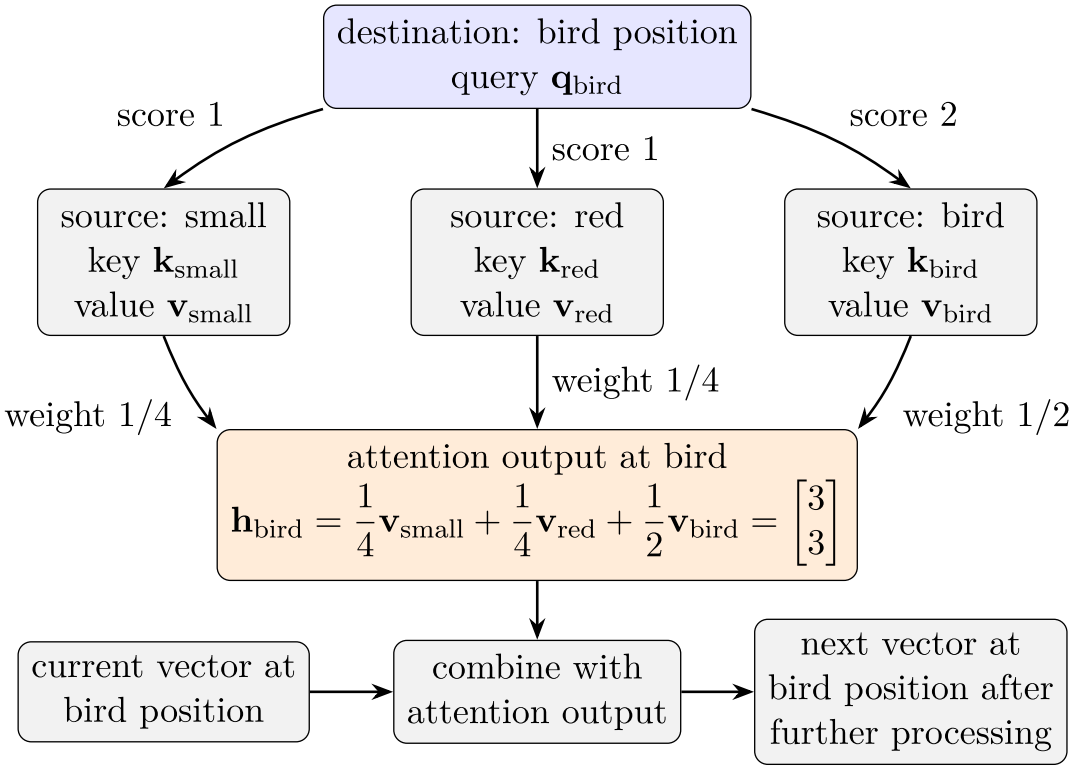


Figure. Attention at the bird position combines value vectors from small, red, and bird.

Warning

This activity isolates one part of a transformer layer. It uses normalization by the sum of positive scores to keep the arithmetic simple. Real transformer attention usually uses a different rule to convert scores into weights, but it follows the same pattern:

$$\text{scores} \longrightarrow \text{weights} \longrightarrow \text{weighted average of values.}$$

A full layer then combines the attention output with the current token representation and processes the result further.